zoom and examine the image, noting the number of discernible pairs of lines that are visible. The more the merrier.

Ergonomics

Ergonomics are a very big factor when it comes to choosing your camera because how you use your camera will have a big impact on your photography. For example, if you're looking for a camera that you're going to use every day, a lightweight camera with well-placed buttons will work wonders and if you're looking for one to stuff in your pocket, obviously, the slimmest camera is the one that you want.

Pick up the camera and take a photo with it. If you're holding the camera correctly, the camera should be stable when taking the shot. Your fingers should be able to easily access the shutter button and the resistance shouldn't be too high. At the same time, check for tactile

feedback when you half-press the shutter to focus. Another thing to note is that when taking a photo is that your fingers shouldn't be covering things like focus assist lamps or that you're almost pressing other buttons that you don't need. Basically, the camera needs to feel good in your hands. If you're looking for a pocket-able camera, put it in your pocket. The next thing you need to look at is the menu and how easy it is to access and set the following options:

- ISO
- Exposure
- White Balance
- Focus Modes
- Shutter Speed
- Aperture Size

extremely small size of the sensors and lenses involved.

Shutter Speed: This is a factor that determines the length of your exposure. The longer the shutter speed, the more the light captured and the brighter the image. The faster the shutter speed, the better for capturing movement. Most good cameras will manage a shutter speed range of 1/2000 – 30 seconds, which is more than enough for your needs.

Aperture and shutter speed go hand-in-hand in determining the quality and type of image that you capture. A lens with a large aperture can use a faster shutter speed, giving you more flexibility in the kind of photos that you can capture.

AutoFocus: The technology behind your camera's AutoFocus (AF) system determines how fast your camera can focus on an object.

- **Phase-detect AF:** This type of AF system is fast and precise. It examines offset images of the same object and uses the amount of offset to determine the precise focus point. This method is predominantly used in DSLRs and is key to their precision and speed.
- **Contrast-Detect AF:** Slow and unreliable, this system uses the image sensor to determine the highest contrast between neighbouring pixels. The problem here is that the sensor will not know if an image is in focus till it passes the point of precise focus at least once.

If you're not taking a DSLR, you rarely have the option of going with a phase-detect system, however, certain types of mirrorless cameras do come with a hybrid AF system (contrast+phase) that considerably hastens the focussing process.

Focal length: This is a figure that you'll find on most camera lenses, even compact digital cameras, and it's usually represented by what's known as a 35mm equivalent figure like 28-135mm. To explain that, consider a lens as something that focuses beams of light to create an image, and in the case of a camera, that image is the size of the image sensor.

Since a 35mm image sensor is considered as a base-line, the 35mm equivalence of a particular lens is the theoretical focal length of that lens if it was designed for the same performance on a 35mm sensor. As an example, 18mm on an APS-C sensor is equivalent to 35mm on a Full-Frame sensor.

There are many ways to define focal length, but the simplest way to put it is that focal length is the distance at which a lens focuses the collimated beams of light. Since 35mm film was the gold standard for photography for quite a while, all focal length today is indicated by its 35mm equivalence.

The focal length determines the field of view (FoV) of the lens, which is effectively, an angle that describes the lens's cone of vision.

- **Wide:** This is an FoV that is very, well, wide and is a term used to describe the FoV of a lens with a focal length below 35mm and such lenses are called wide-angle lenses.

- **Normal:** A lens with a focal length of 35-50mm is generally considered to have a "normal" FoV, in other words, an FoV that very closely resembles the natural FoV of the human eye, and such lenses are called normal lenses.
- **Narrow:** This FoV is best described by a lens with a focal length in excess of 50m and such lenses are called telephoto lenses.

Zoom lenses are lenses that have the ability to traverse a range of focal lengths, no matter how small, so something that goes from 10mm to 20mm is also a zoom lens.

Prime lenses are lenses with a fixed focal length. These lenses are usually much cheaper than zoom lenses and offer stellar image quality. d